

Eduardo Aguilar-Bejarano

Machine Learning for Asymmetric Catalysis & Molecular Design

PhD, Chemistry & Artificial Intelligence · University of Nottingham

✉ ed.aguilar.bejarano@gmail.com 📞 +44 7835 267035 🌐 edaguilarb.github.io

🌐 EdAguilarB 🆔 0009-0007-6251-2350 🎓 Google Scholar

PROFILE

Chemist working on machine learning for asymmetric and organometallic catalysis. My research addresses how chiral transition-metal complexes are represented for predictive modeling, and how interpretable models can be turned into testable structure–selectivity hypotheses for catalyst design — developed throughout in close collaboration with synthetic groups, with predictions validated prospectively on newly synthesized catalysts and substrates. I also work on generative models for molecular design, explainability methods for graph neural networks, the benchmarking of chemical foundation models, and open-source scientific software for experimental researchers.

Interests

Asymmetric & organometallic catalysis · representation of chiral metal complexes · interpretable graph neural networks · generative molecular design · benchmarking & foundation models · open-source scientific software

EDUCATION

2022 – 2026

PhD in Chemistry & Artificial Intelligence

Nottingham, UK

University of Nottingham — AI Doctoral Training Centre

Thesis: Interpretable Graph Neural Networks for the Discovery of Structure–Activity Relationships

Supervisors: Prof. Simon Woodward, Prof. Graziela Figueredo, Prof. Ender Özcan ·

Defended 7 September 2026

2018 – 2022

BSc (Hons) Chemistry

San José, Costa Rica

Universidad de Costa Rica

GPA 9.23 / 10.0 · Graduated with distinction (academic excellence)

Relevant coursework: Bio-organic Chemistry, Bio-inorganic Chemistry, Computational Tools for Physical Chemistry, Scientific Computation with Python, Introduction to Genomics and Metagenomics Data Processing, Automated Learning with Python.

RESEARCH EXPERIENCE

Nov 2024 –
Present

Data Scientist (part-time)

Nottingham, UK

Digital Research Service, University of Nottingham

EPSRC Large Grant: Biomaterials for Medical Devices AI

- Lead modeler for an antibiofilm polymer-discovery program: trained ensemble GNNs on ~500 polymer structures against high-throughput biofilm assays, in iterative collaboration with experimental microbiologists.
- Delivered **LCNet**, the neural-network model behind the first quantitative Q-Cryo-OrbiSIMS method for lipid-nanoparticle characterization (with MIT's Anderson Lab); [ChemRxiv preprint](#).
- Architect and delivery lead for **Helix**, an open-source framework for reproducible, interpretable ML on tabular scientific data, adopted across chemistry, materials, and biology at Nottingham.

Jun – Aug 2026

Visiting Researcher

Cambridge, MA, USA

Anderson Lab, Massachusetts Institute of Technology

Generative and interpretable models for molecular design

- Developed conditional generative models (diffusion and VAE latent-space optimization) for the design of novel lipid structures, coupling generative sampling with property-prediction models to steer candidates toward target profiles.
- Built interpretable models to prioritize lipid nanoparticles, extending prior composition-analysis work toward model-guided exploration of chemical space.

Oct 2022 –
Sep 2026

PhD Researcher, Chemistry & Artificial Intelligence

Nottingham, UK

University of Nottingham

AI Doctoral Training Centre · Woodward, Figueredo & Özcan Research Groups

- Showed that graph encoding of planar and axial chirality predicts enantioselectivity as accurately as models built on expert-derived descriptors, without requiring them, and transfers to unrelated reaction classes (catalytic asymmetric dearomatization, N,S-acetal formation).
- Prospective validation with synthetic collaborators: predictions tested against 52 newly synthesized Rh-catalyzed 1,4-addition reactions (Lam Group, Nottingham) and 19 new Pd-catalyzed DAAA reactions (Guiry Group, University College Dublin).
- Established that scaffold-based splits, not random splits, predict the real-world error of asymmetric-catalysis models — a benchmark across fingerprints, descriptors, pretrained chemical foundation models, and GNNs.
- Showed that GNN-based data-quality workflows can flag mislabeled entries in published asymmetric-catalysis datasets.
- Developed a 3D-masking explainability method for physics-based (crystal-convolution) GNNs predicting interstitial-alloy properties, and trained machine-learned interatomic potentials.

- Built classical ML models (multiple linear regression, SVM, random forest, gradient boosting) to predict partition coefficients (logP, water/toluene) and tautomerization constants of drug-like small molecules, including database curation, molecular-representation design, and feature selection.

PUBLICATIONS

Complete, up-to-date list on [Google Scholar](#) and [ORCID](#).

PEER-REVIEWED JOURNAL ARTICLES

- [5] **Aguilar-Bejarano, E.**; Lea, D.; Sivakumar, K.; Mase, J. M.; Omidvar, R.; Li, R.; Kettle, T.; Mitchell-White, J.; Alexander, M. R.; Winkler, D. A.; Figueredo, G. Helix 1.0: An Open-Source Framework for Reproducible and Interpretable Machine Learning on Tabular Scientific Data. *Patterns* **2026**, 7 (5), 101536.
[doi:10.1016/j.patter.2026.101536](#) · IF 10.8
- [4] **Aguilar-Bejarano, E.**; Galvin, D.; Rogers, D. M.; Özcan, E.; Woodward, S.; Guiry, P. J.; Figueredo, G. Benchmarking Molecular Representations and Machine Learning Algorithms for Asymmetric Catalysis: A Palladium-Catalysed Decarboxylative Asymmetric Allylic Alkylation Case Study. *J. Cheminform.* **2026**, 18 (1), 120.
[doi:10.1186/s13321-026-01236-z](#) · IF 7.9
- [3] **Aguilar-Bejarano, E.**; Arrieta, L.; Gutiérrez, M.; Özcan, E.; Woodward, S.; Figueredo, G.; Borge, J. I. Explainable GNN-Derived Structure–Property Relationships in Interstitial-Alloy Materials. *Phys. Chem. Chem. Phys.* **2025**, 27 (41), 22240–22250.
[doi:10.1039/D5CP02208H](#) · IF 3.0
- [2] **Aguilar-Bejarano, E.**; Özcan, E.; Rit, R. K.; Li, H.; Lam, H. W.; Moore, J. C.; Woodward, S.; Figueredo, G. Homogeneous Catalyst Graph Neural Network: A Human-Interpretable Graph Neural Network Tool for Ligand Optimization in Asymmetric Catalysis. *iScience* **2025**, 28 (3), 111881.
[doi:10.1016/j.isci.2025.111881](#) · IF 4.5
- [1] **Aguilar-Bejarano, E.**; Deorukhkar, V.; Woodward, S. Data Checking of Asymmetric Catalysis Literature Using a Graph Neural Network Approach. *Molecules* **2025**, 30 (2), 355.
[doi:10.3390/molecules30020355](#) · IF 5.1

PREPRINTS

- [P1] Roberts, J. W.; Kotowska, A.; **Aguilar-Bejarano, E.**; Collins, E.; Witten, J.; Howard, A.; Hook, A.; Winkler, D. A.; Alvey, K.; Figueredo, G.; Scurr, D. J.; Anderson, D. G.; Alexander, M. R. Quantitative Cryogenic Orbitrap Secondary Ion Mass Spectrometry for Structural Characterization of Lipid Nanoparticles. *ChemRxiv* **2026**.
doi:10.26434/chemrxiv.15001097

OPEN-SOURCE SOFTWARE

- Catalysis **HCat-GNet** — interpretable GNN for ligand and catalyst optimization in asymmetric catalysis; attributes predictions to chemically meaningful substructures.
 [EdAguilarB/hcatgnet](#) · Paper [2]
- Catalysis **AsymBench** — benchmarking suite for asymmetric catalysis; modular base classes for adding representations, models, and extrapolative splits.
 [EdAguilarB/AsymBench](#) · Paper [4]
- Polymers **PolyNet** — polymer property prediction: six GNN architectures, automated hyperparameter optimization, atom- and fragment-level explainability, and a GUI for non-computational users.
 [Biomaterials-for-Medical-Devices-AI/PolyNet](#)
- General ML **Helix** — reproducible, interpretable ML on tabular scientific data. Architect and delivery lead, including interpretation modules, GUI, and documentation. Adopted across chemistry, materials, and biology at Nottingham.
 [Biomaterials-for-Medical-Devices-AI/Helix](#) · Paper [5]
- Lipid NPs **LCNet / LNPQuantification** — neural-network model behind the first quantitative Q-Cryo-OrbiSIMS method for lipid-nanoparticle characterization (with MIT's Anderson Lab); Integrated-Gradients interpretation.
 [Biomaterials-for-Medical-Devices-AI/LipidLNPQuantification](#) · Preprint [P1]
- Materials **InterstitialAlloys** — explainable GNNs for interstitial-alloy property prediction.
 [EdAguilarB/InterstitialAlloys](#) · Paper [3]

HONOURS & AWARDS

FELLOWSHIPS & SCHOLARSHIPS

- 2022 **AI Doctoral Training Centre Scholarship**, University of Nottingham
Full four-year PhD funding (tuition and stipend) through a competitive EPSRC programme.
- 2022 **CeNAT-CONARE Research Fellowship**, National Center for High Technology, Costa Rica
US\$4,300 for the project “Cheminformatic and Quantum Mechanics Approaches for Quantitative Prediction of Tautomerism in Bioactive Molecules”.
- 2021 **GCURS Acceptance & Travel Award**, Rice University, USA
- 2019 – 2022 **Academic Excellence Scholarship**, Universidad de Costa Rica
Annual award for a prior-year GPA of 9.0/10 or above.

PRIZES & DISTINCTIONS

- 2024 **Christopher J. Moody Synthesis & Catalysis Poster Prize**, School of Chemistry, University of Nottingham
- 2024 **Second Place, PGR Symposium Poster Contest**, University of Nottingham
- 2022 **Bachelor of Science with Honours**, Universidad de Costa Rica
Awarded for graduation with academic excellence.
- 2022 **Best Scientific Poster**, Chemistry Students Association, Universidad de Costa Rica

OLYMPIADS

- 2021 **Gold Medal**, Ibero-American University Organic Chemistry Olympiad, Universidad Nacional de Colombia
- 2017 **Gold Medal**, Costa Rican Physics Olympiad; **Honourable Mention**, Costa Rican Chemistry Olympiad, Universidad Nacional (UNA)

CONFERENCES & PRESENTATIONS

INVITED TALKS

- Mar 2024 *HCat-GNet: An Interpretable Graph Neural Network for Catalysis Optimization*
The GSK Prosperity Partnership Stevenage, UK
- Aug 2023 *Chemistry in the Artificial Intelligence Era*
School of Chemistry Seminar Series, Universidad de Costa Rica
San José, Costa Rica

CONTRIBUTED TALKS

- Sep 2023 *Feature Identification in Molybdenum Carbides: Graph Neural Networks vs. Human Empirical Search* (oral and poster)
Machine Learning for Atomistic Modelling Autumn School, Daresbury Laboratory
Daresbury, UK
- Oct 2021 *Cheminformatic and Quantum Mechanics Approaches for Quantitative Prediction of Tautomerism in Bioactive Molecules*
Gulf Coast Undergraduate Research Symposium (GCURS), Rice University
Houston, TX, USA

POSTERS

- Jul 2024 *HCat-GNet: an Interpretable Graph Neural Network for Catalysis Optimization*
School of Chemistry Postgraduate Symposium, University of Nottingham
Nottingham, UK
- Nov 2023 *Feature Identification in Molybdenum Carbides: Graph Neural Networks vs. Human Empirical Search*
Faraday Community Poster Symposium, Burlington House
London, UK
- Sep 2022 *Development of a Novel Lipophilicity Descriptor from Toluene/Water Partition Coefficient Predictions*
Sciences Week Scientific Poster Contest
San José, Costa Rica
- Apr 2022 *Machine Learning Methods to Determine the Toluene/Water Partition Coefficient as an Efficient Lipophilic Descriptor*
University Week Scientific Poster Contest
San José, Costa Rica

SUPERVISION & TEACHING

RESEARCH SUPERVISION

- 2024, 2026 **Co-supervisor, summer research interns** · Woodward & Figueredo Groups, University of Nottingham
- 2026: projects on prediction error / applicability-domain estimation and the open-source Helix tool.
 - 2024: GNN-based curation of asymmetric-catalysis datasets (resulting in *Molecules* 2025, paper [1]).
- 2023 – 2024 **Co-supervisor, MSc Machine Learning in Science** · University of Nottingham
- 2024: three students, with Prof. Simon Woodward.
 - 2023: one student (Veera Pedireddi), with Prof. Graziela Figueredo and Dr. Kristaps Ermanis.
- Deep-learning projects in the chemical sciences.

TRAINING & WORKSHOPS

Nottingham Designed and delivered in-house Python and machine-learning training sessions for interdisciplinary research teams.

TEACHING & TUTORING, COSTA RICA

Jul – Sep 2022 **Programming, ML & Data-Analysis Tutor**, CBIO3 Group
Teaching assistant for Prof. William Zamora's Cheminformatics course.

Sep – Nov 2021 **Organic & Analytical Chemistry Tutor**, Campus Atlántico

Mar – Oct 2021 **Chemistry Olympiad Tutor**, Colegio Científico Costarricense, San Pedro

2021, 2022 **General Chemistry Tutor** for incoming students (March intakes)

2019 – 2021 **Laboratory Teaching Assistant** — seven courses
Physical Chemistry I & II, Analytical Chemistry I, Organic Chemistry I, General Organic Chemistry I, Chemical Experimentation II, General & Qualitative Chemistry.

Mar – Jul 2019 **Physics Reinforcement Tutor**, Colegio Científico Costarricense, San Pedro

2018 **Teaching Assistant** to Prof. Darío Chinchilla (General & Biological Chemistry)

PROFESSIONAL SERVICE & MEMBERSHIPS

Peer review Reviewer for *Journal of Cheminformatics* (Springer Nature), *Chemometrics and Intelligent Laboratory Systems* (Elsevier), and *Archives of Computational Methods in Engineering* (Springer Nature).

2023 – Present Royal Society of Chemistry — Associate Member

2021 – 2022 American Chemical Society — Member

2021 – 2022 Costa Rica ACS Student Chapter — Member

TECHNICAL SKILLS

Cheminformatics & Molecular Modeling

RDKit · SMILES/InChI · molecular representations (fingerprints, topological & physicochemical descriptors, molecular graphs, organometallic complexes) · QSAR/QSPR · Chemprop · OpenBabel · DataWarrior · PyMOL

Deep Learning & Machine Learning

PyTorch · PyTorch Geometric · GNNs (GCN, GAT, MPNN, GraphSAGE) · generative models (diffusion/flow-based, VAEs, latent-space optimization) · scikit-learn · XGBoost · transfer & self-supervised learning · ensemble methods · hyperparameter optimization (Ray Tune)

Interpretability & Foundation Models

GNExplainer · SHAP · Integrated Gradients · GradCAM · attention analysis · chemical foundation models (ChemBERTa, MolT5, UniMol) · LLM APIs (Anthropic Claude, OpenAI GPT)

Atomistic Modeling & Interatomic Potentials

Machine-learned interatomic potentials (MACE) · ASE · pymatgen · DScribe · quantum chemistry (Gaussian — QM continuum solvation)

Software Engineering & Computing

Python (advanced, OOP) · Git & code review · automated testing · CI/CD · Docker · Bash · R · Linux · HPC/SLURM · GPU/CUDA · Weights & Biases · Streamlit

Laboratory

HPLC · gas chromatography · atomic absorption · UV/Vis spectrophotometry · fluorescence spectrophotometry

Languages

Spanish (native) · English (fluent)

REFEREES

Prof. Simon Woodward — PhD primary supervisor

School of Chemistry, University of Nottingham

simon.woodward@nottingham.ac.uk

Prof. Graziela Figueredo — PhD co-supervisor & line manager

Digital Research Service, University of Nottingham

g.figueredo@nottingham.ac.uk

Prof. Ender Özcan — PhD co-supervisor

School of Computer Science, University of Nottingham

ender.ozcan@nottingham.ac.uk